I Measurements :-

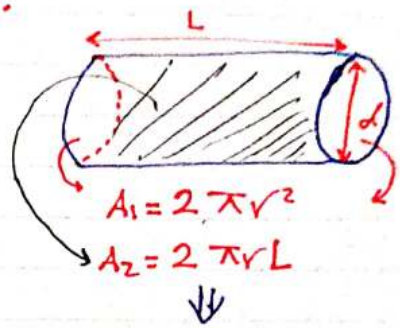
$r = \frac{d}{2}$

* objectives:- To study the Vernier scale principle and to lern the use of Vernier callipers for the accurate measurement of length. To become familiar with the use of micrometer callipers for the accurate measurement of small lengths.

* Cylinder :-

* $A = 2\pi r L + 2\pi r^2$

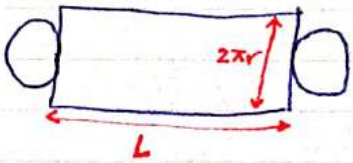
نسبة الخطأ $\frac{\Delta A}{A} = \frac{\Delta r}{r} + \frac{\Delta L}{L} + 2\frac{\Delta r}{r}$



* $V = \pi r^2 L = \frac{\pi d^2 L}{4}$

نسبة الخطأ $\frac{\Delta V}{V} = 2\frac{\Delta r}{r} + \frac{\Delta L}{L}$

$\frac{\Delta V}{V} = 2\frac{\Delta d}{d} + \frac{\Delta L}{L}$



* Spherical :-



* $A = 4\pi r^2 = \pi d^2$

نسبة الخطأ $\frac{\Delta A}{A} = 2\frac{\Delta r}{r} = 2\frac{\Delta d}{d}$

* $V = \frac{4}{3}\pi r^3 = \frac{\pi}{6} \cdot d^3$

نسبة الخطأ $\frac{\Delta V}{V} = 3\frac{\Delta r}{r} = 3\frac{\Delta d}{d}$

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⊛ Orthogonal parallelepiped:-



⊛ $V = L.w.h$

$$\frac{\Delta V}{V} = \frac{\Delta L}{L} + \frac{\Delta w}{w} + \frac{\Delta h}{h}$$

⊛ $A_{of\ an\ face} = L.h \;/\; A = w.h \;/\; A = w.h$

$$\frac{\Delta A}{A} = \frac{\Delta L}{L} + \frac{\Delta h}{h} \;/\; \frac{\Delta A}{A} = \frac{\Delta w}{w} + \frac{\Delta h}{h} \;/\; \frac{\Delta A}{A} = \frac{\Delta w}{w} + \frac{\Delta h}{h}$$

⊛ Density :-

$$\rho = \frac{m}{V} \Rightarrow \text{Error :-}\ \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

⊛ Caliper / الورنية :- 0.05 mm
⊛ Ruler / المسطرة :- 0.5 mm
⊛ Micrometer :- 0.005 mm

⊛ $\frac{|tru - exp|}{tru} * 100\%$ → نسبة الخطأ المئوية

⊛ $\frac{|slop - tru|}{tru}$ → نسبة الخطأ القياسية

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[2] Vectors :-

⊛ objectives :- The objective of this lab is add vectors using both the tail-to-head method and the component method and to verify the results using a force table.

⊛ $F = M.g$

⊛ $|\vec{R}| = \sqrt{R_x^2 + R_y^2} \longrightarrow \theta = \tan^{-1}\left(\frac{\Sigma F_x}{\Sigma F_y}\right)$

⊛ ~~$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta_{12}}$~~ ⊛ Cos law $= \sqrt{F_1^2 + F_2^2 + |2F_1F_2\cos\theta_{12}|}$

⊛ Sin law $= \frac{\sin B}{B} = \frac{\sin A}{A} = \frac{\sin C}{C}$

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[3] Newton's Second law:-

- objectives:- * Proof experimentally that the relation between the net force and acceleration is directly proportional. (When the total mass is constant).

  * Proof experimentally that the relation between the acceleration and the total mass in inversely proportional. (When the net force is constant).

- $F = ma$
- $F_{net} = m_h \cdot g$
- $a = \frac{2.5}{t^2}$
- $F_{net} = (m_h + M_w) \cdot a$ ($m_h \cdot g$)
- $slop = \frac{\Delta F}{\Delta a}$

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[4] Uniform Circular Motion:-

- objectives:- * Alows examining the relation between mass, velocity, radius, and centripetal force.

  * To experimentally measure the magnitude of the rubber bob mass.

- $F_c = Mg$
- $F_c = \frac{4\pi^2 r m}{T^2}$ $\left(F_c = \left(\frac{m}{r}\right) v^2\right)$
- $v = \frac{2\pi r}{T}$
- $a_c = \frac{v^2}{a}$
- $f = \frac{1}{T}$

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5 The Simple Pendulum :-

⊛ objectives:- * Determine an experimental value of the acceleration due to gravity (g).

* Investigate the dependence of the Period(T) of a pendulum on the length (L) and the mass(m) of the bob.

⊛ $T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow T^2 = \frac{4\pi^2 L}{g} \Rightarrow g = \frac{4\pi^2 L}{T^2}$

6 Simple Harmonic Motion (الحركة التوافقية البسيطة) :-

⊛ objectives:- * Determine the spring constant (K) of a spring by measuring the elongation versus applied force.

* Determine the spring constant (K) of a spring from measuring of the period of oscillation.

* Investigate the dependence of the period(T) of oscillation of a mass on the value of the mass and on the amplitude of the motion.

⊛ $K = \frac{F}{\Delta x}$ ⊛ $\sigma = \sqrt{\frac{\Sigma(K_i - \bar{K})^2}{n-1}}$ ⊛ $K = \frac{4\pi^2 M}{T^2}$

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7 Gravitationl potential energy:-

⊛ objectives:- * Find the acceleration of gravity experimentally:-
1- Fixing (hi) and change (m).
2- Fixing (m) and change (hi).

⊛ $K = m.g.h_i$ ⊛ $v = L/t$ ⊛ $K = 1/2 (m+M)v^2$

⊛ $K_f + U_f = K_i + U_i$

$U_f = K_i = Zero$, $K_f = U_i$

8 One-Dimensional Collisions:-

⊛ objectives:- * To verify the law of conservation of linear momentum in one-dimensional collisions.

* To study conservation of energy and linear momentum in both elastic an inelastic one-dimensional collisions.

⊛ $P = mv$ ⊛ $KE = 1/2 mv^2$ ($_{total}$) ⊛ $v = L/T$

* inelastic:- $KE_{before} \neq KE_{after}$

* elastic:- $KE_{before} = KE_{after}$

KOMEKS

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